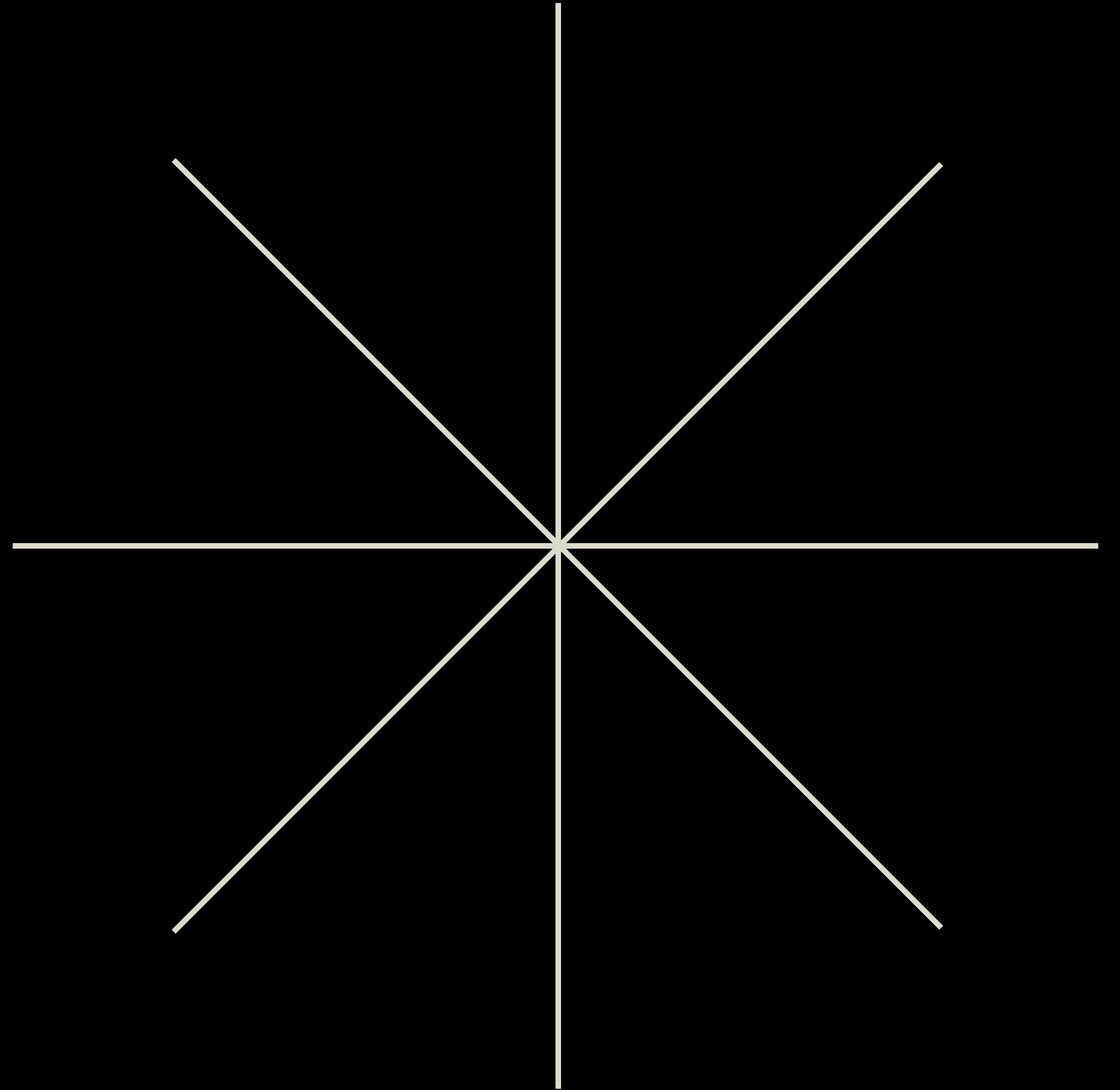
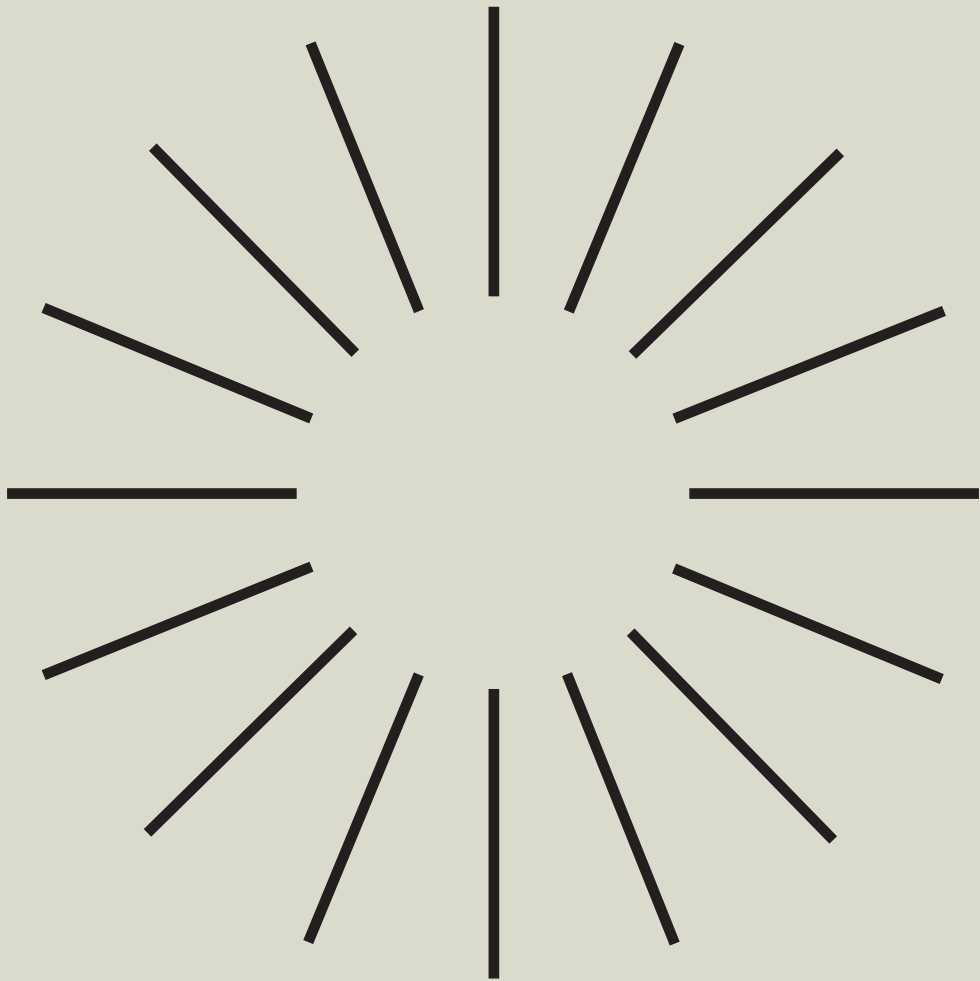


Kaleidoscopic reorganization of network communities across different scales

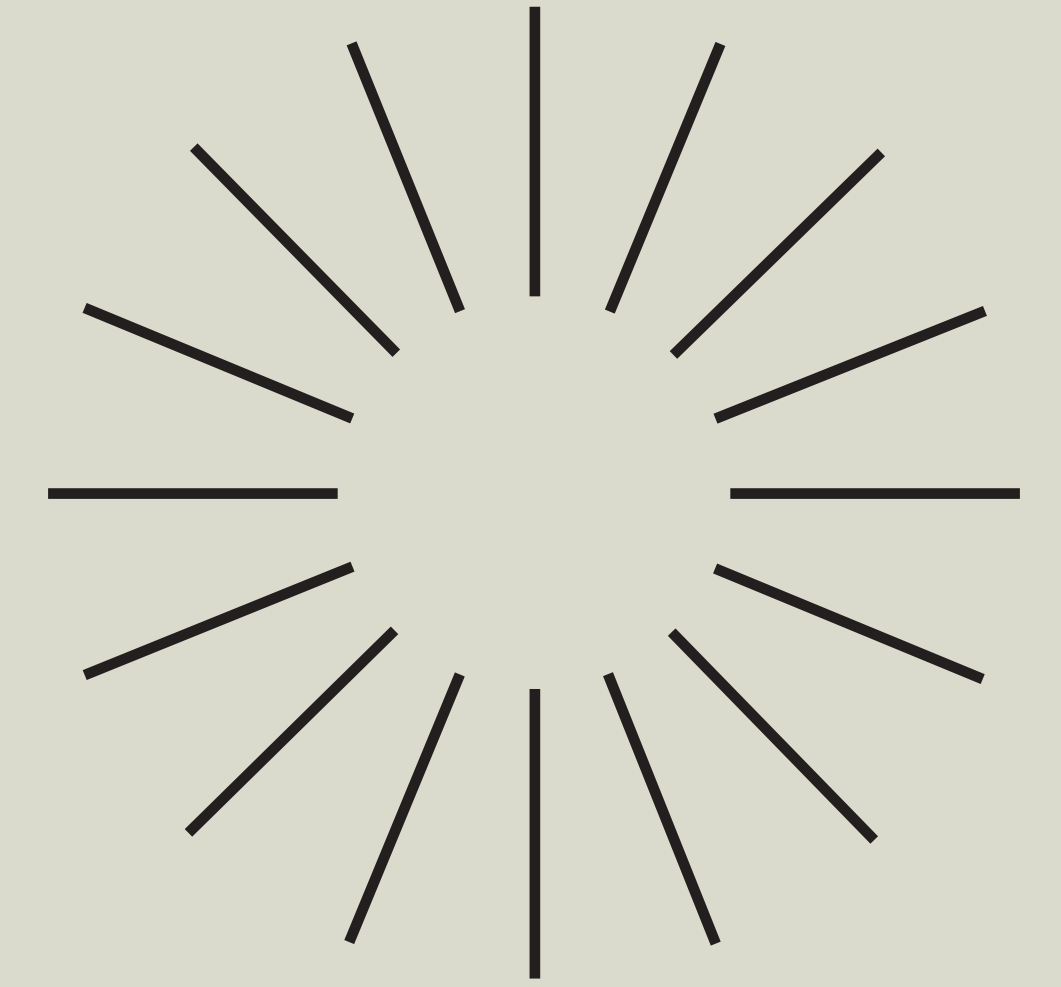


MODULARITY



$$Q(G;\gamma)=\frac{1}{M}\sum_g\left(L_g-\gamma\frac{K_g^2}{4M}\right)$$

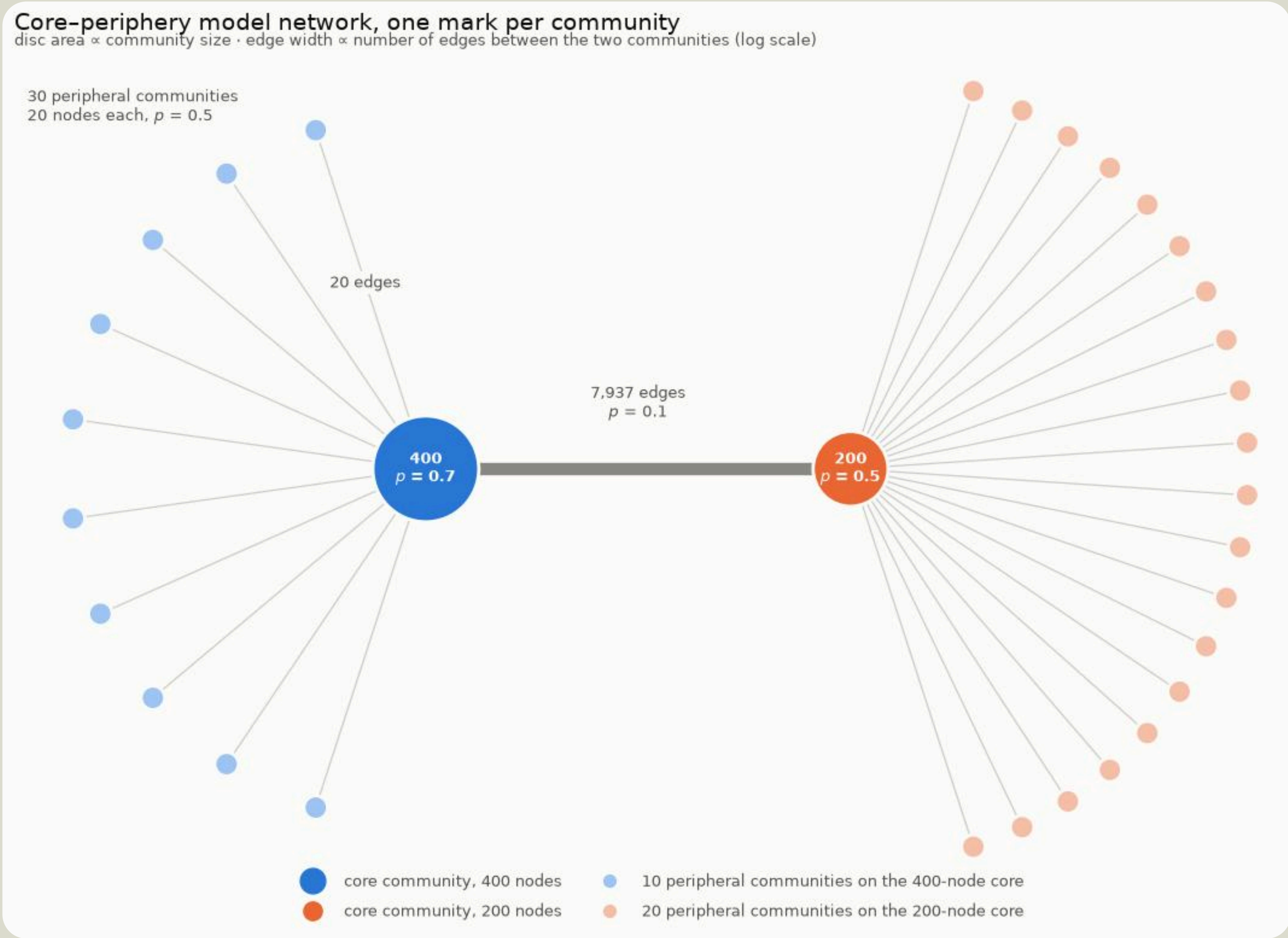
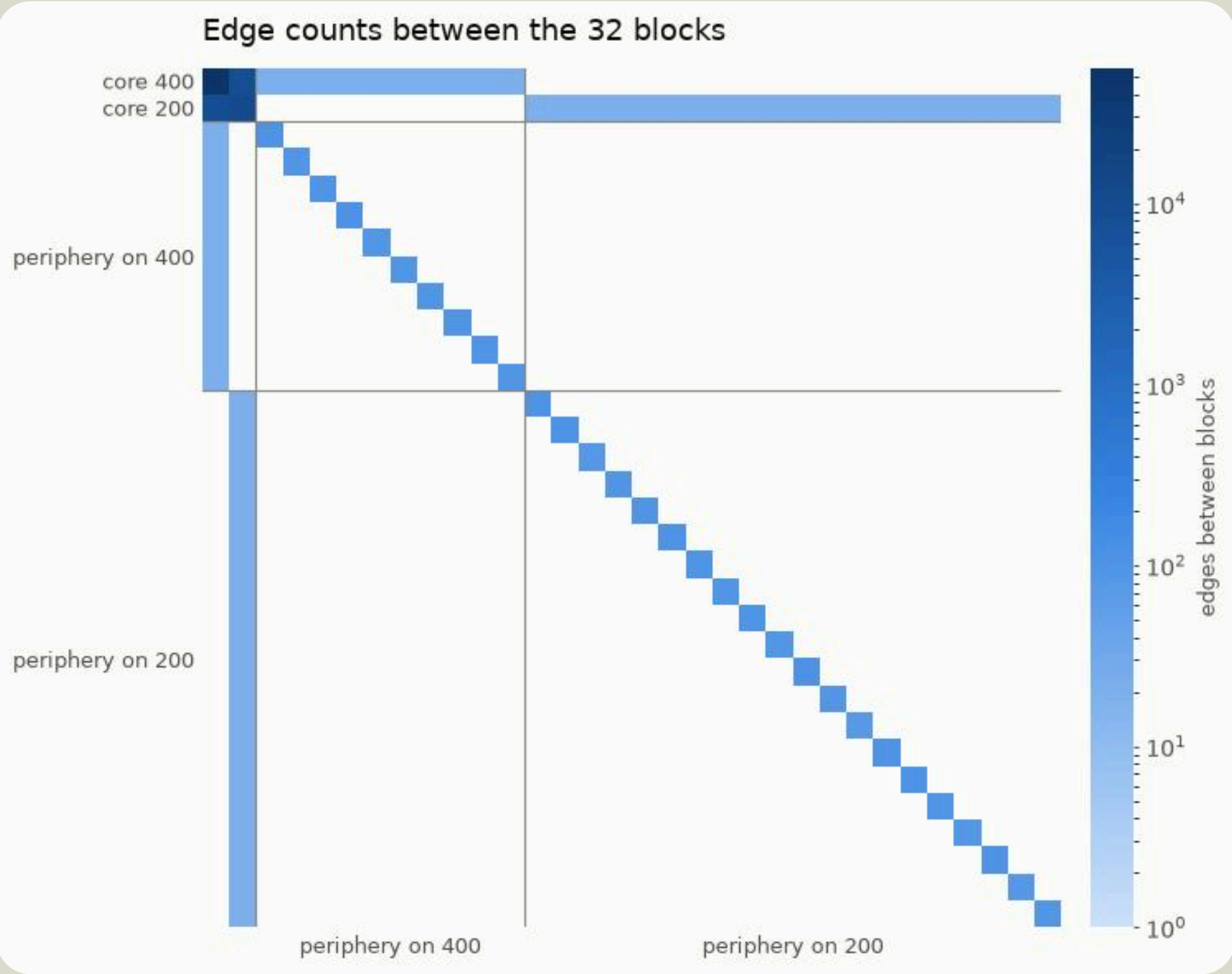
MODULARITY



$$Q(G; \gamma) = \frac{1}{M} \sum_g \left(L_g - \gamma \frac{K_g^2}{4M} \right)$$

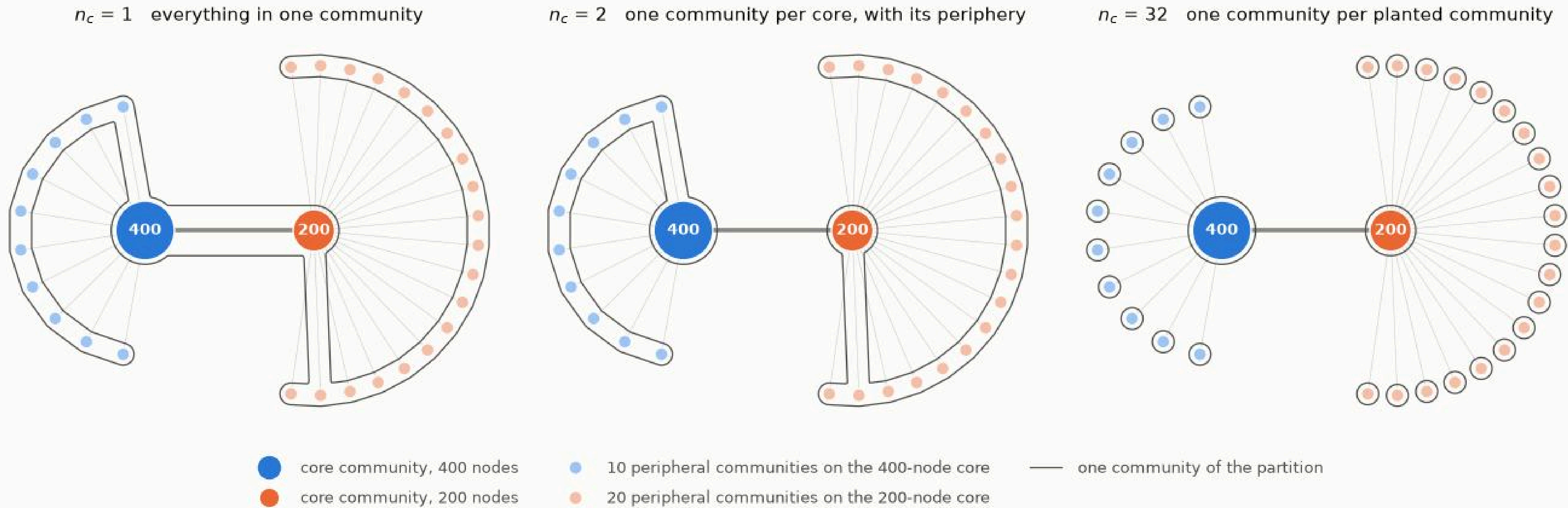
```
1 G = NX.KARATE_CLUB_GRAPH()  
2 GAMMA = 1.0  
3 COMMUNITIES = LOUVAIN_COMMUNITIES(G, RESOLUTION=GAMMA, SEED=42)  
4 Q = MODULARITY(G, COMMUNITIES, RESOLUTION=GAMMA)
```

RESOLUTION

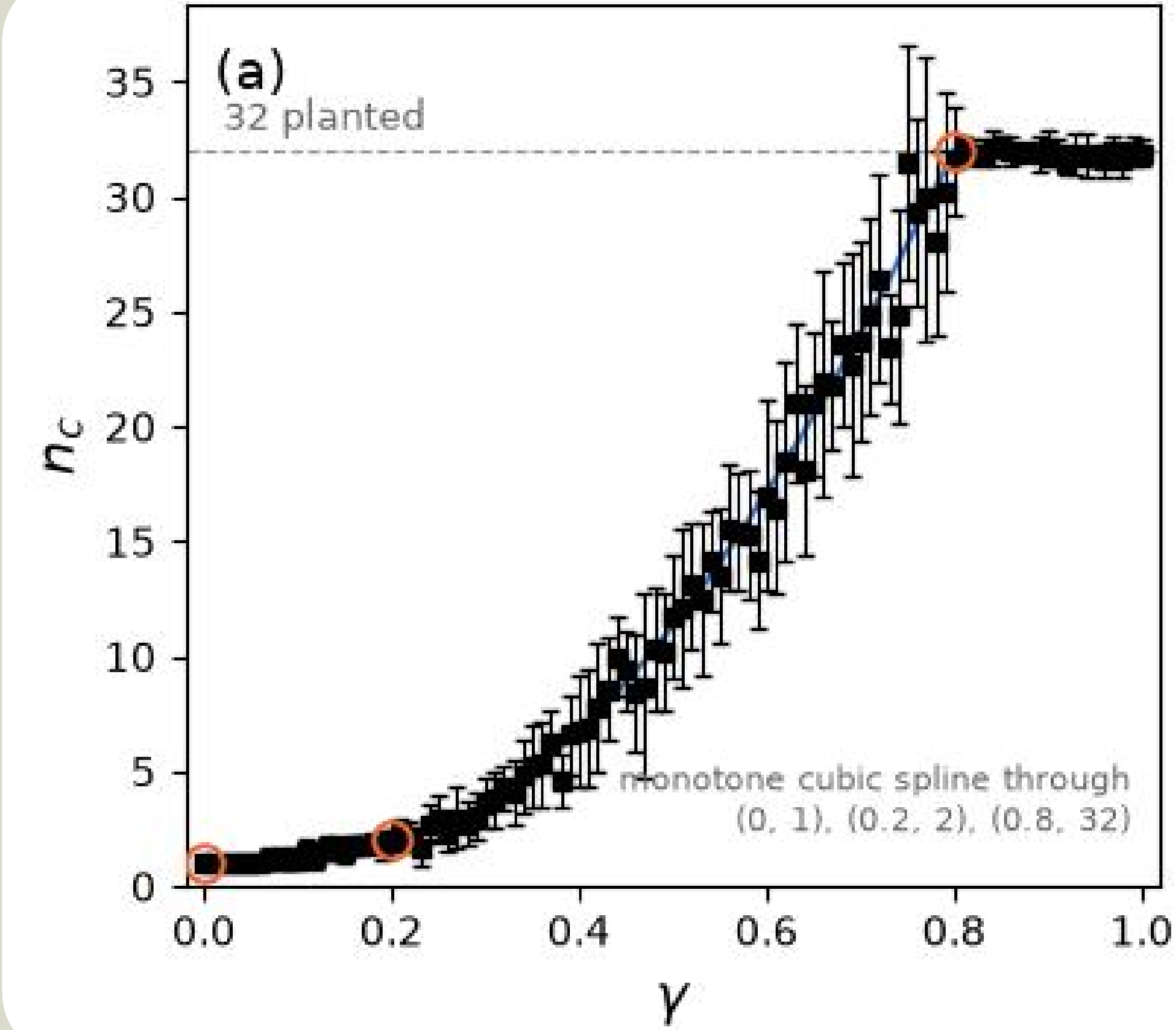
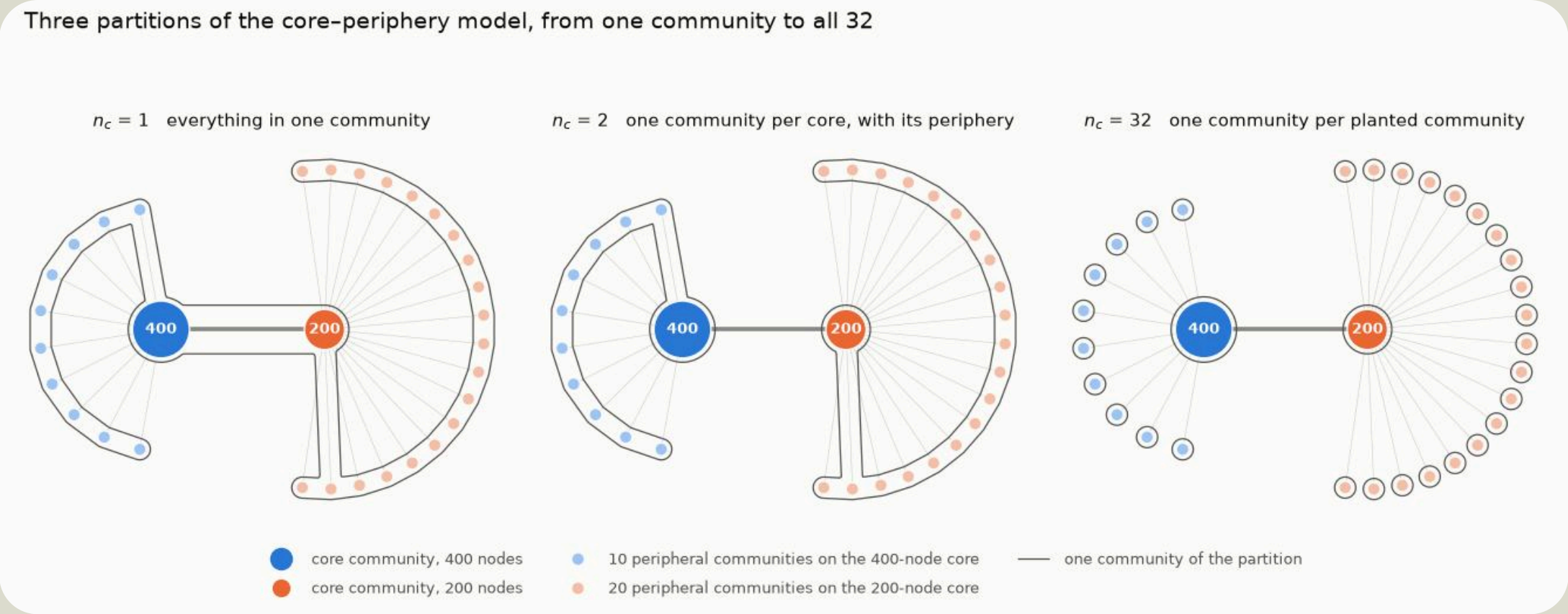


RESOLUTION

Three partitions of the core-periphery model, from one community to all 32

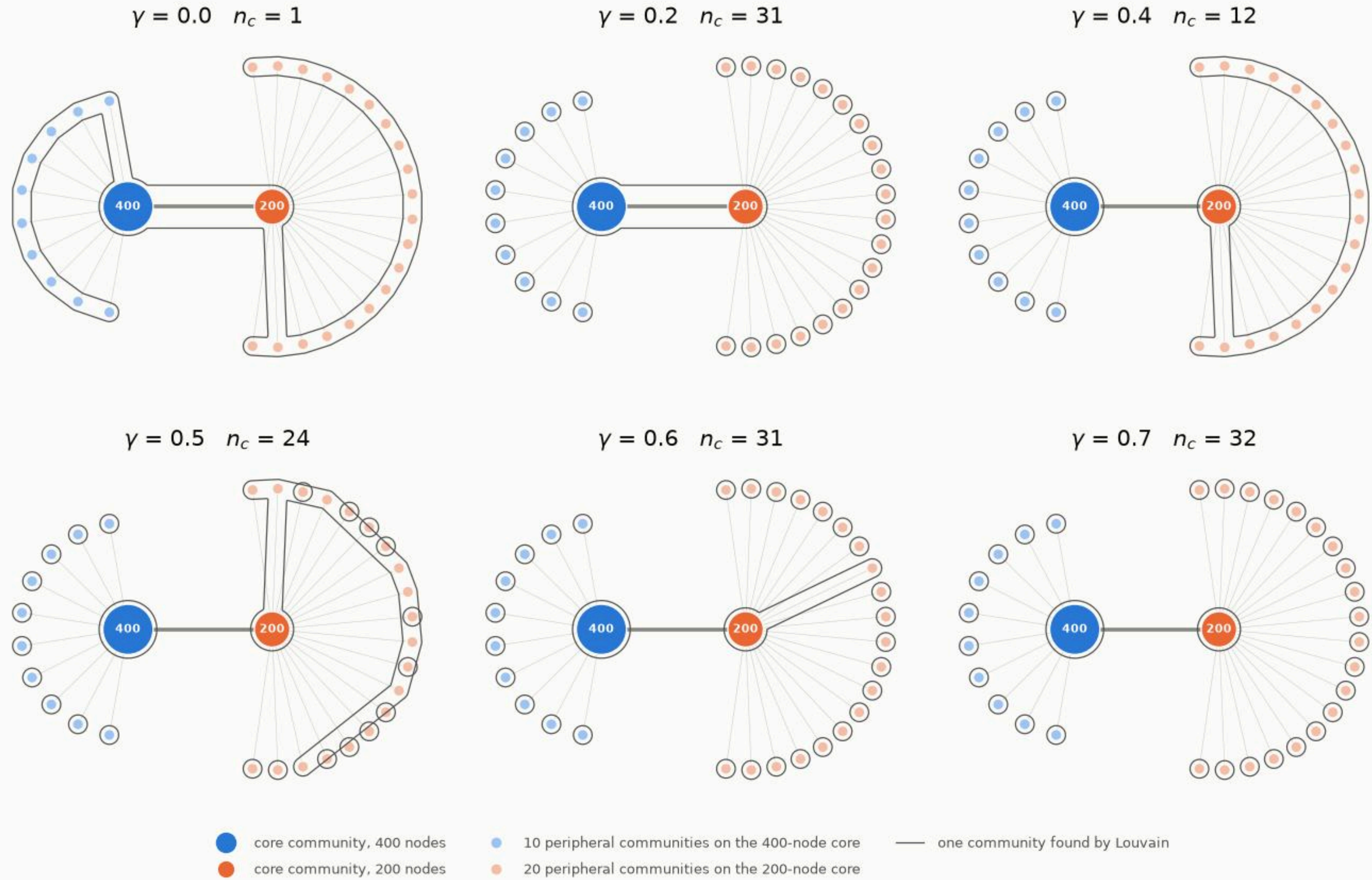


RESOLUTION

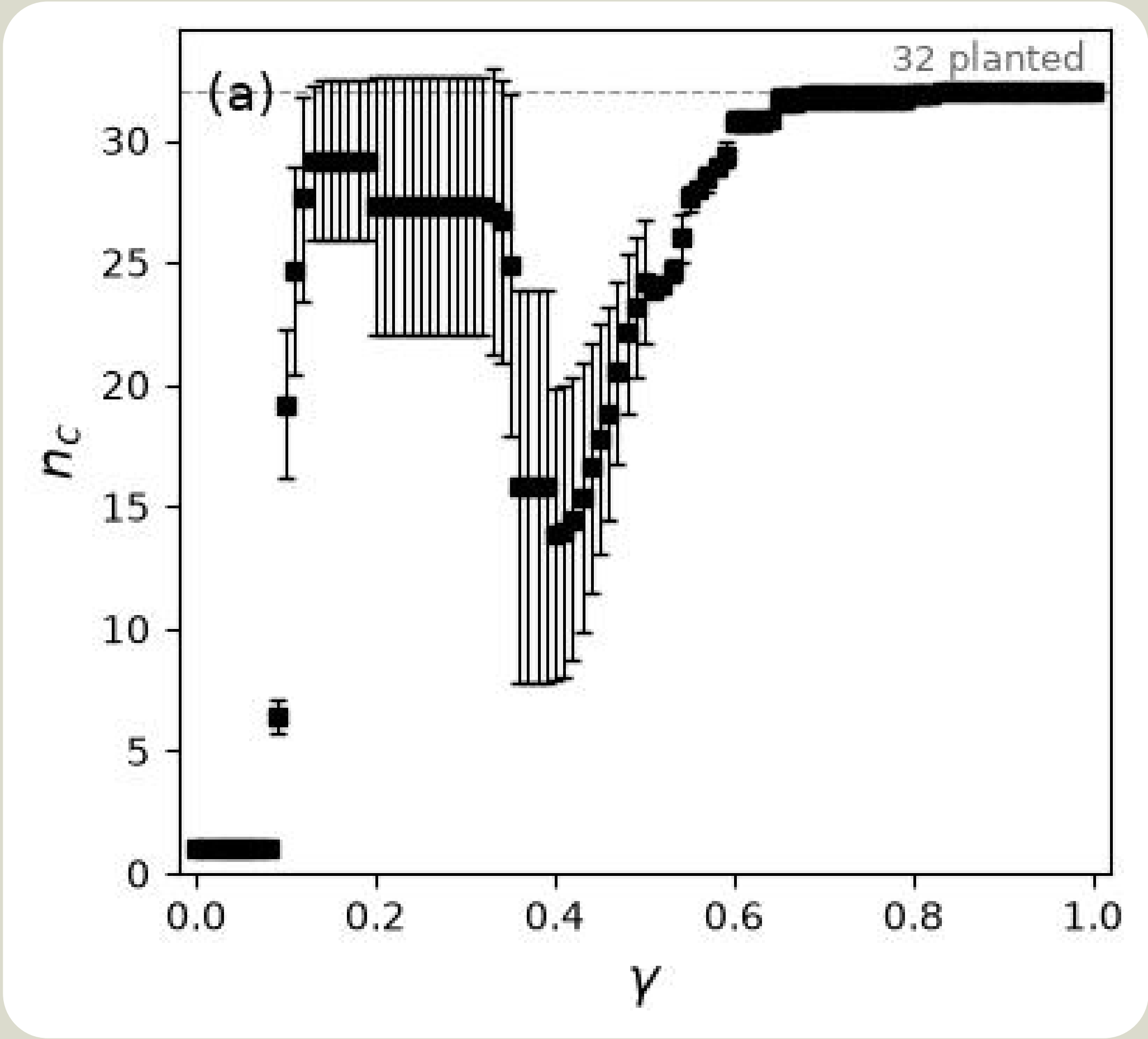
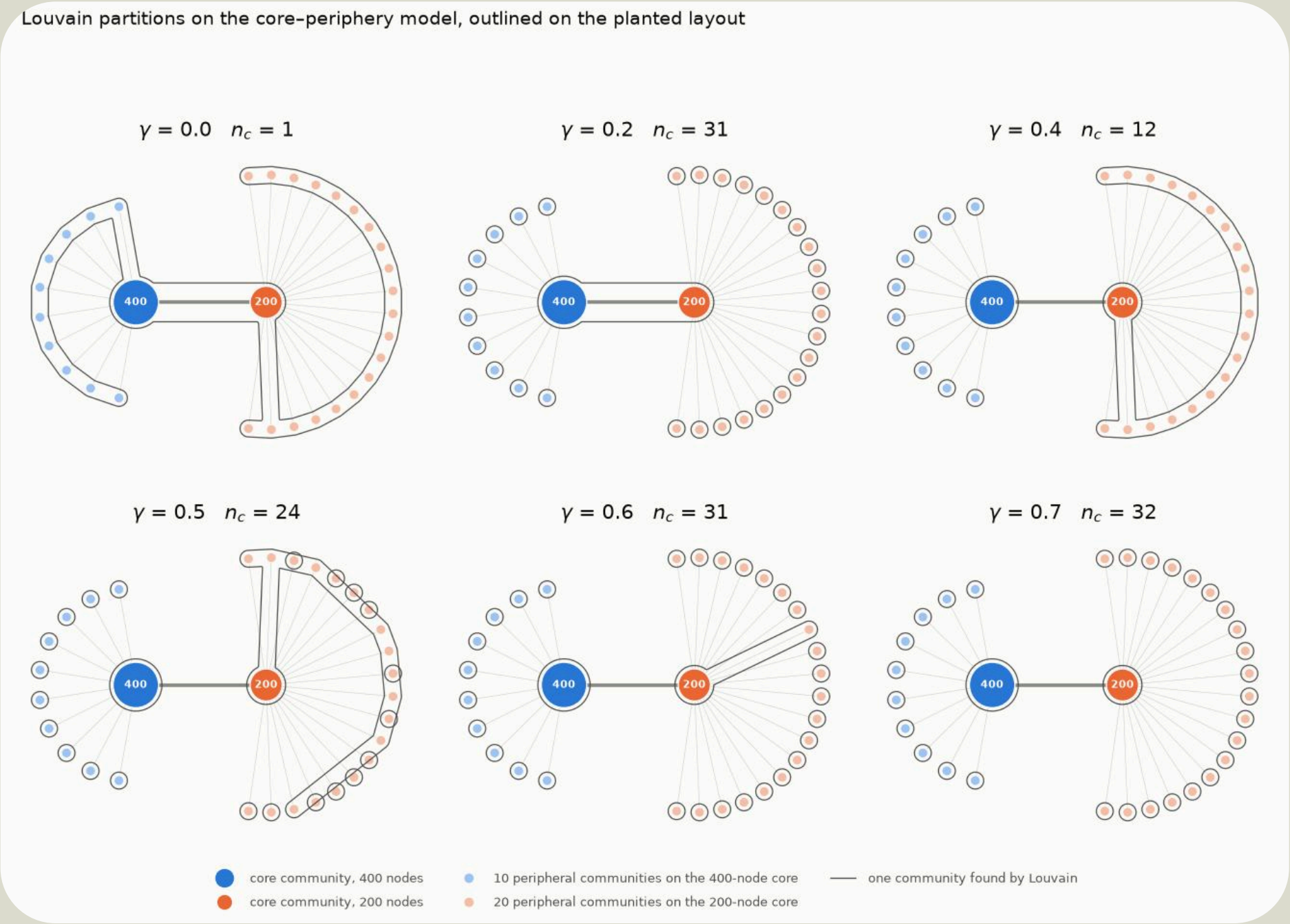


BEHAVIOR

Louvain partitions on the core-periphery model, outlined on the planted layout



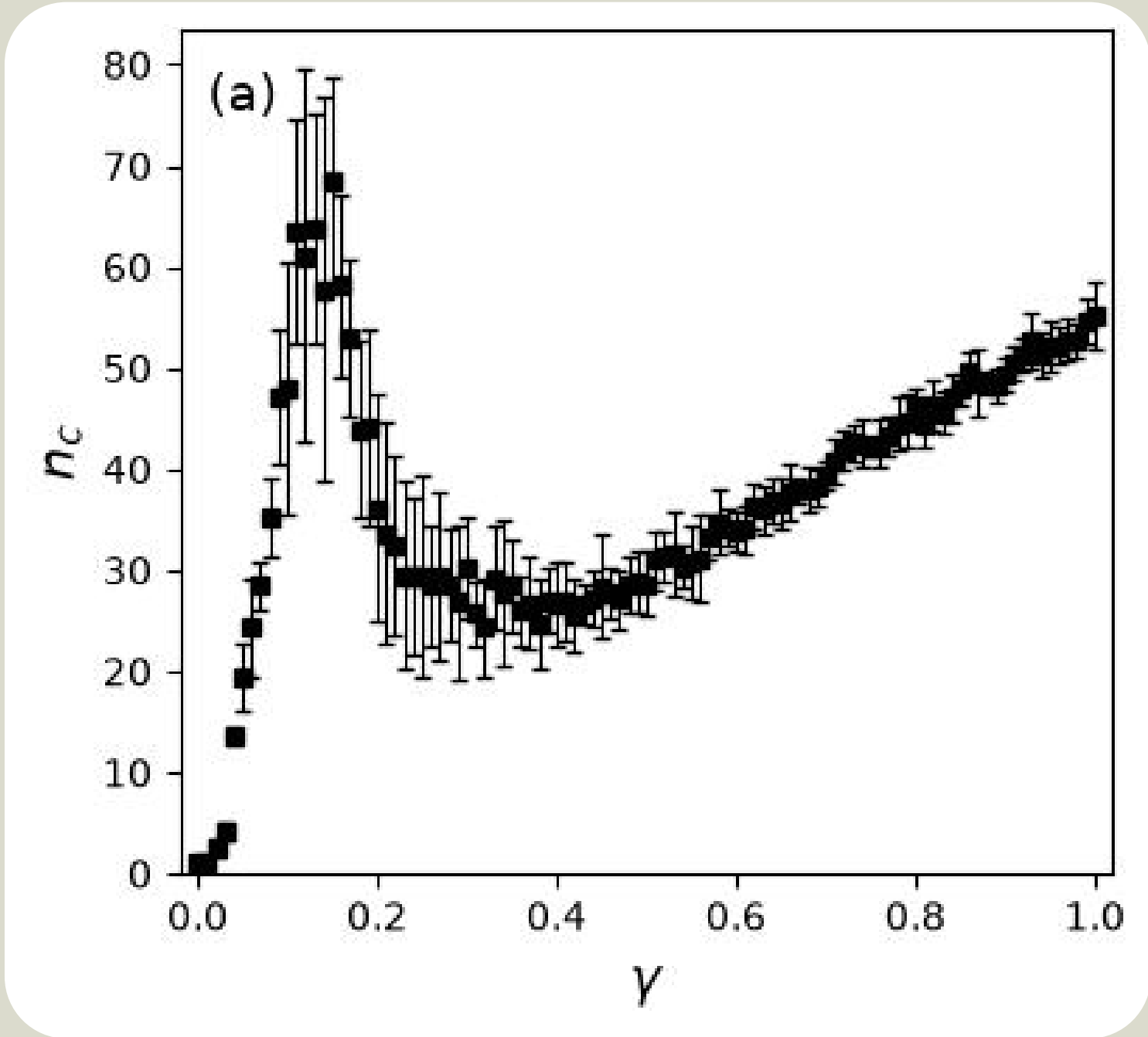
BEHAVIOR



REAL NETWORKS

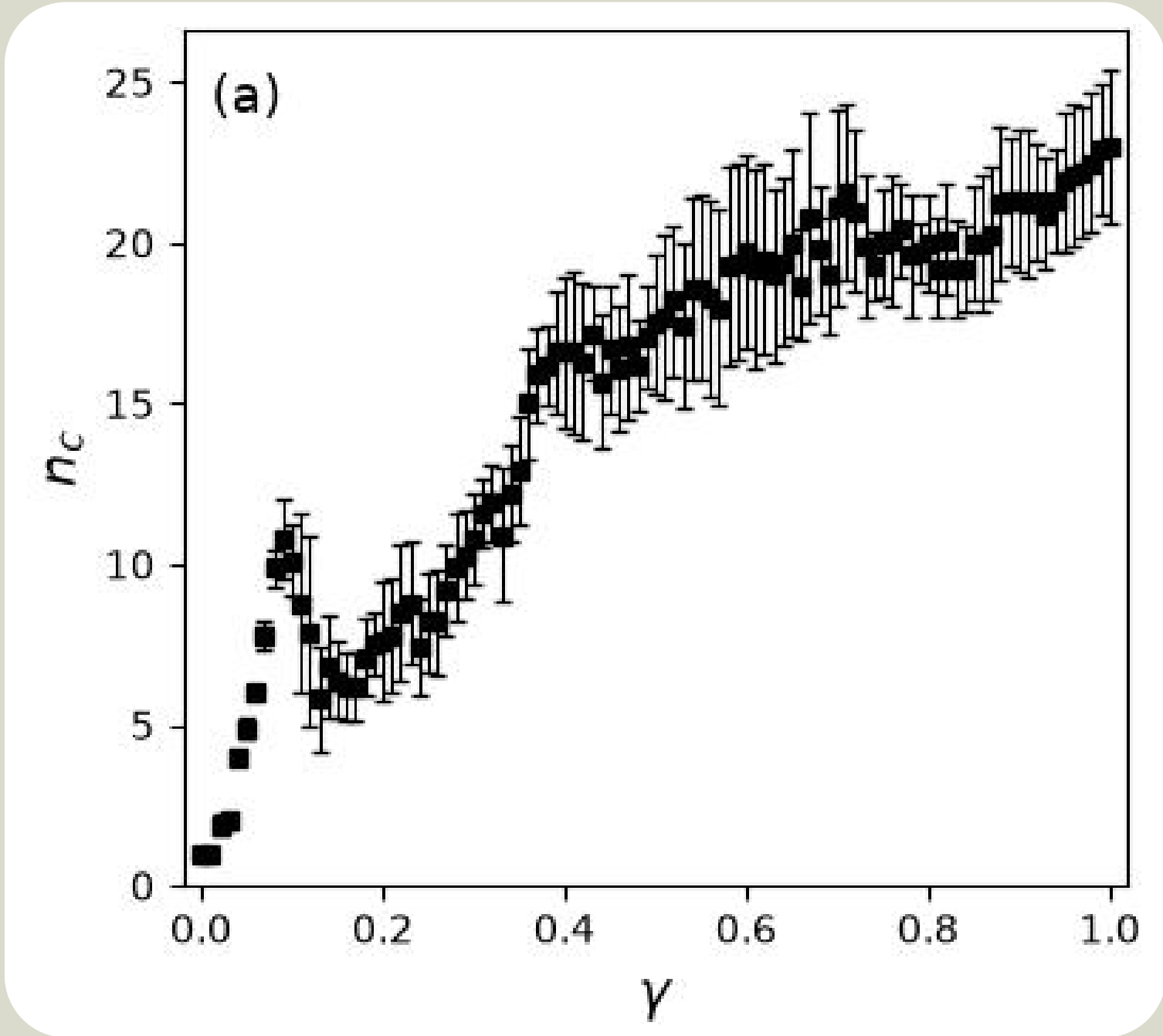
CondMat

scientific collaboration network of
co-authors in condensed matter physics



openFlights - flights

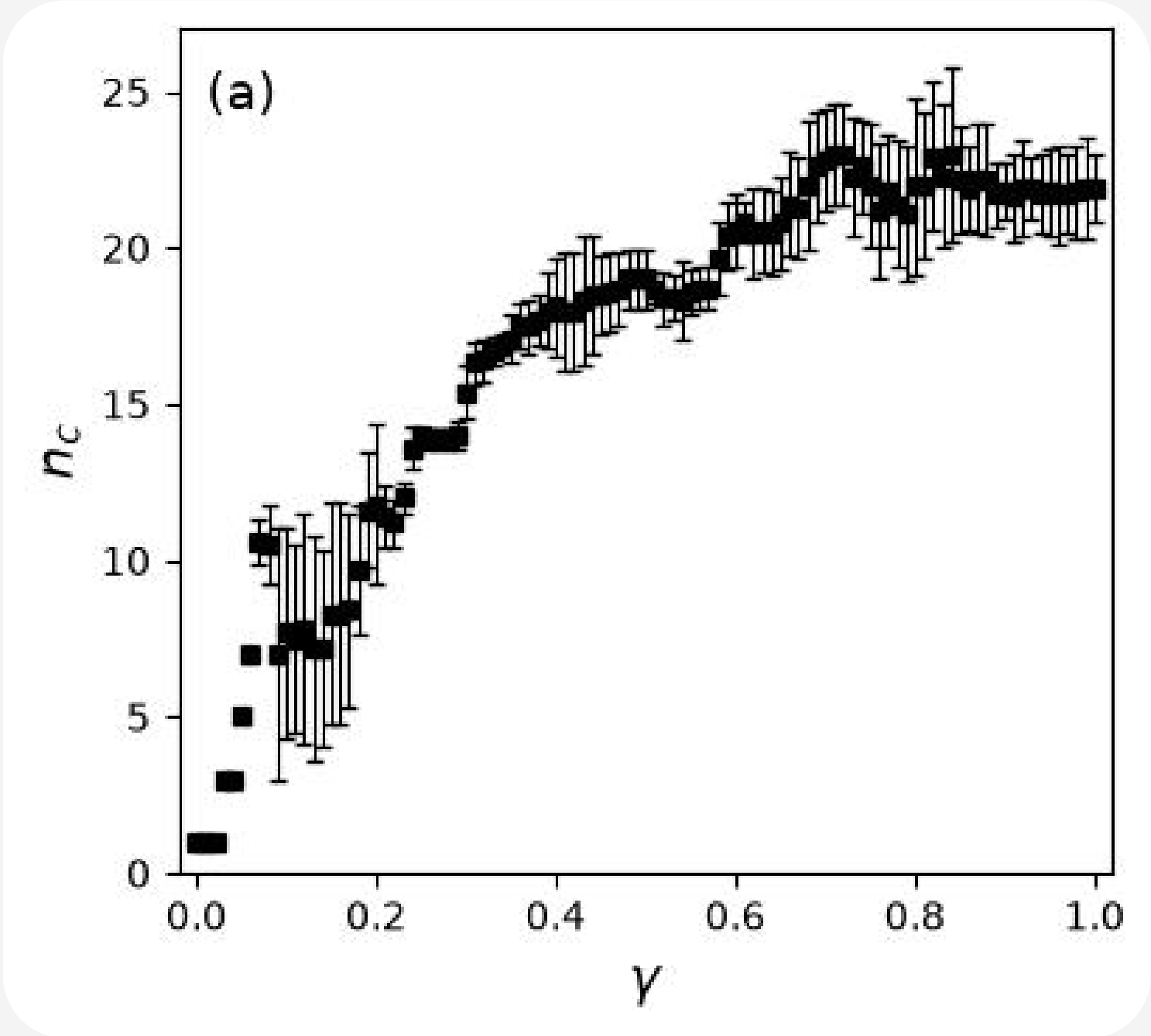
frequency of daily travel along specific
routes between different airports



(*1) - NOT THE RULE

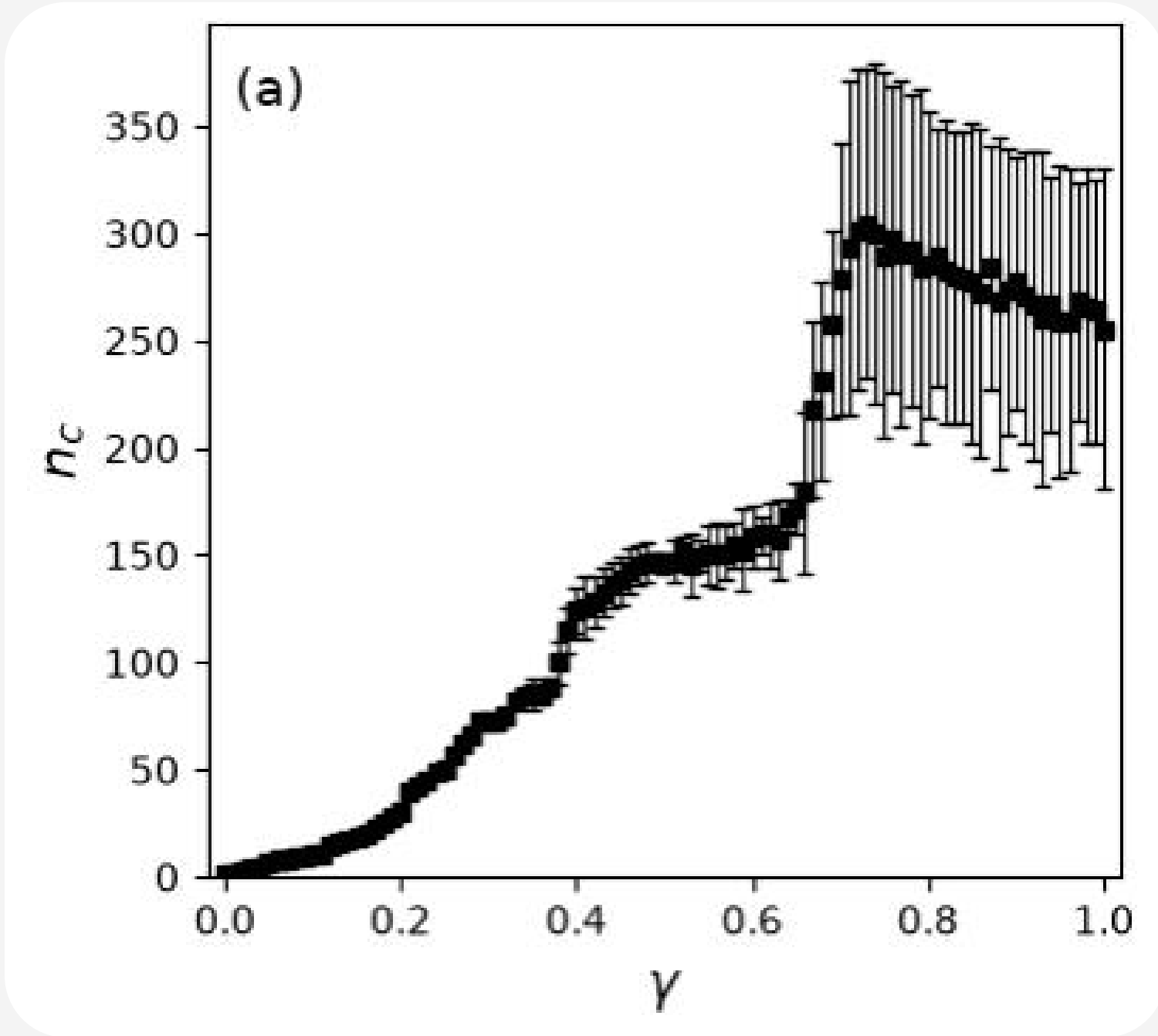
rt-pol

retweets on US politics

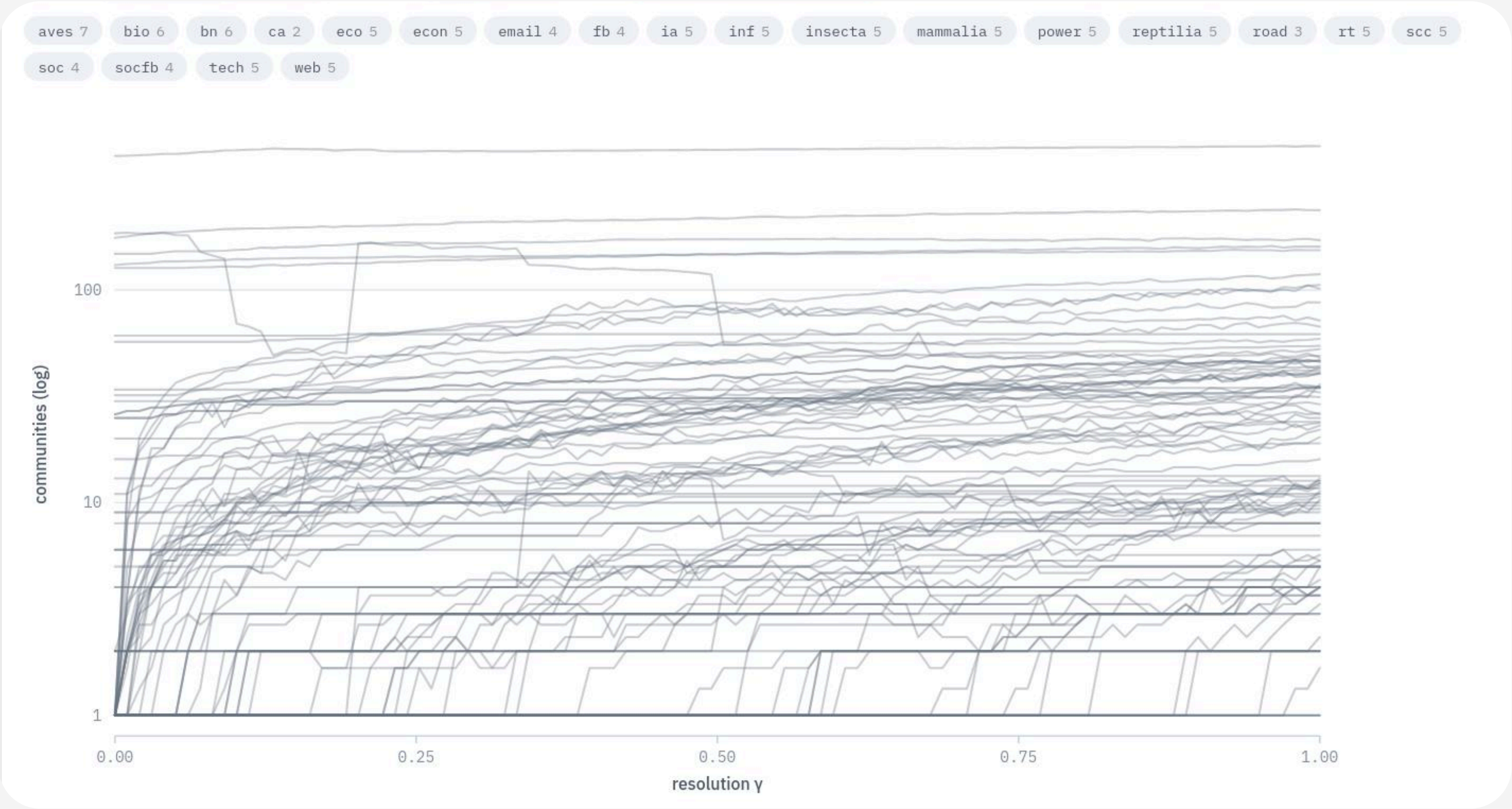


openFlights - routes

direct zero-stop structural connections
between different airport nodes



(*1) - NOT THE RULE

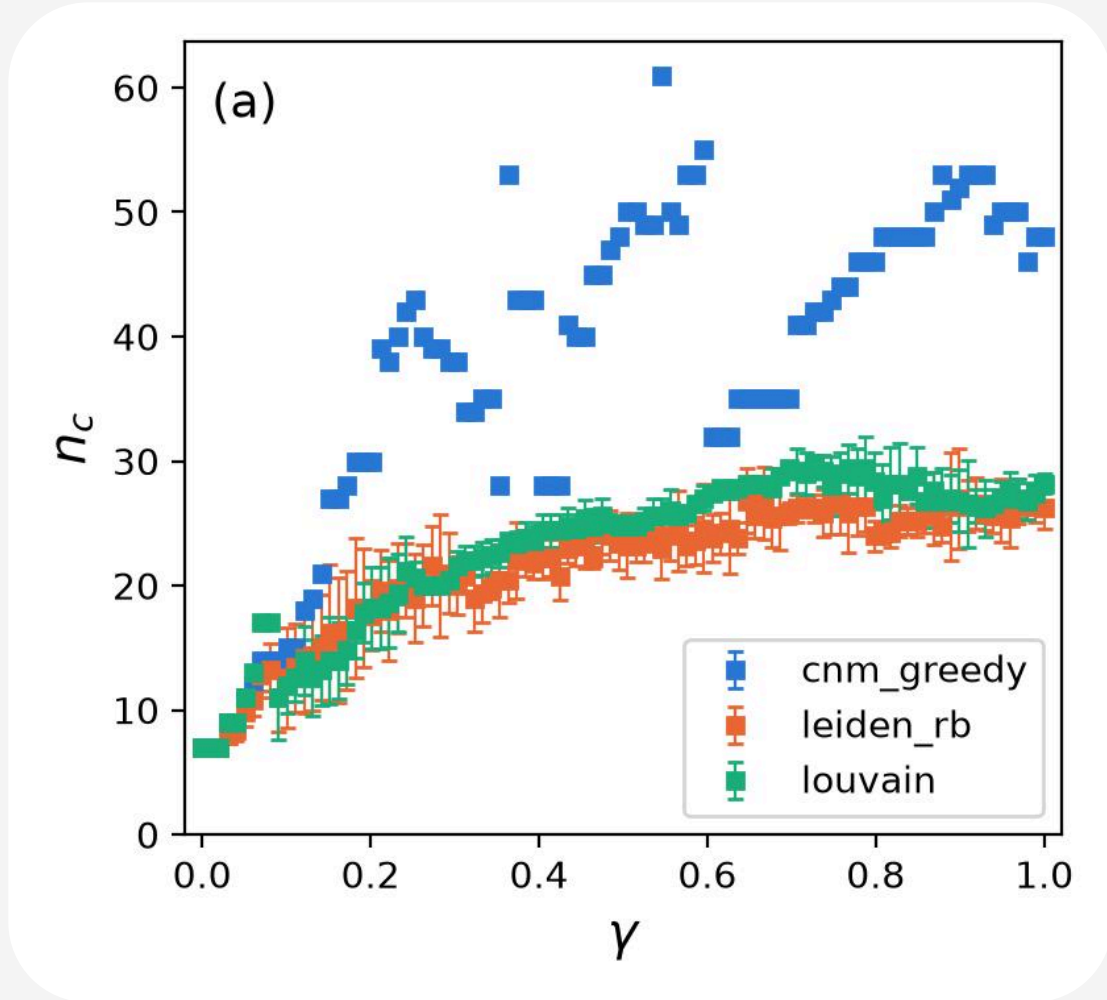
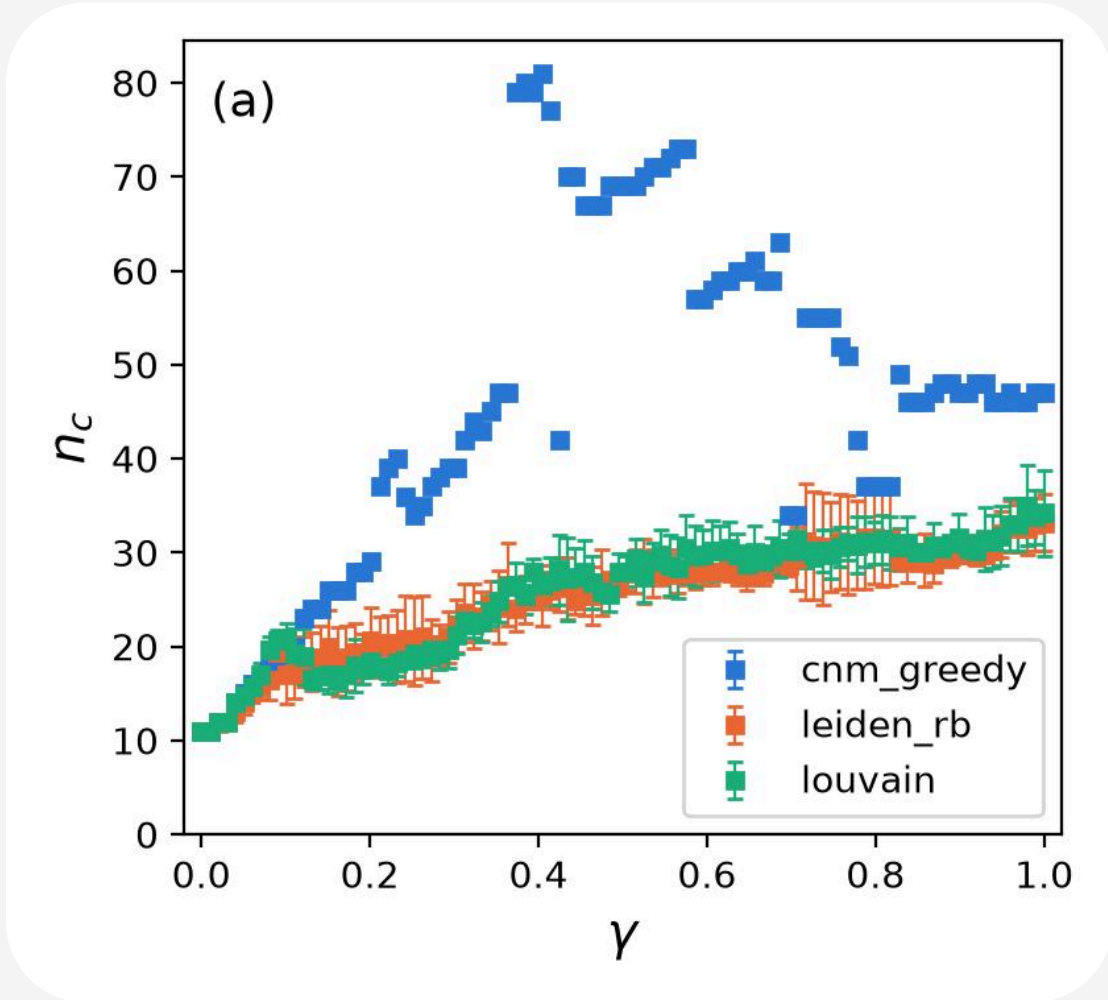
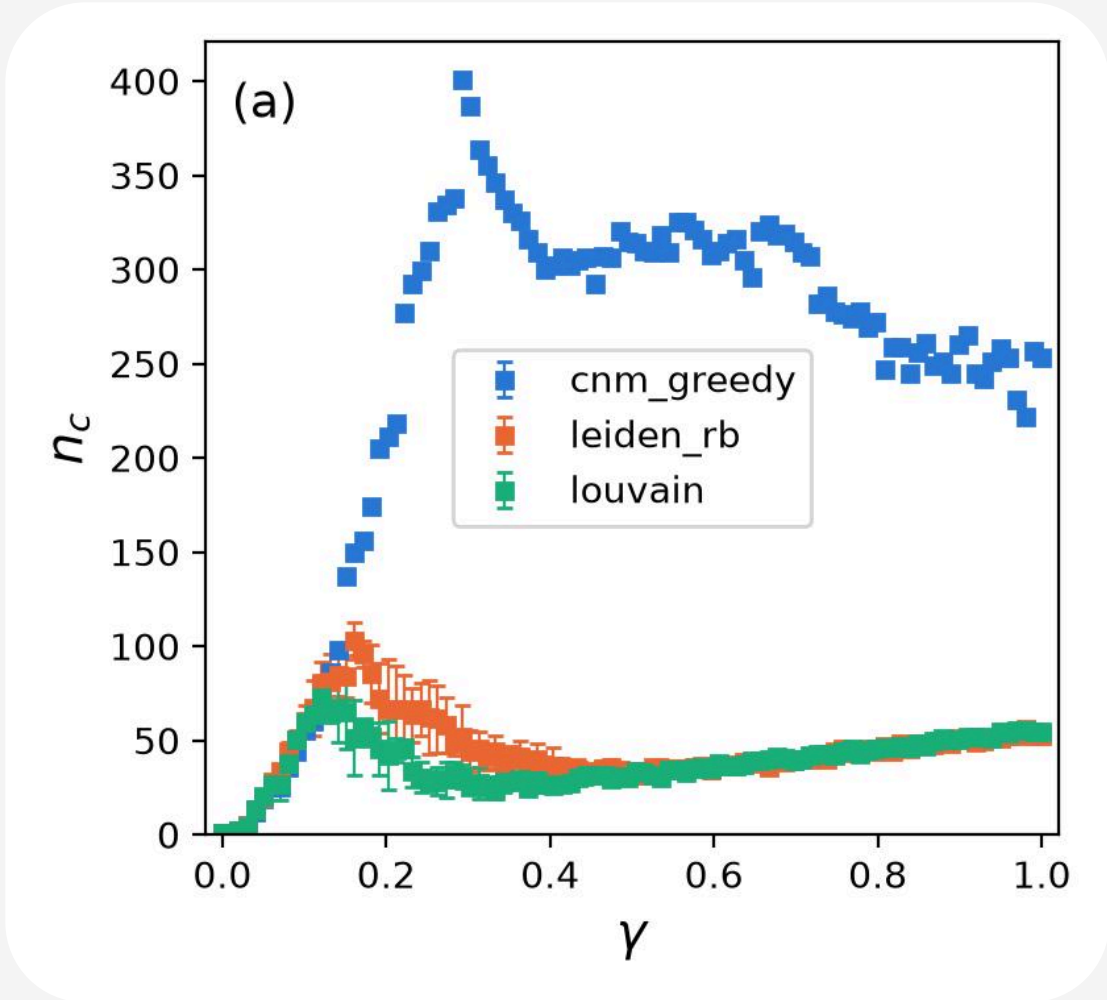
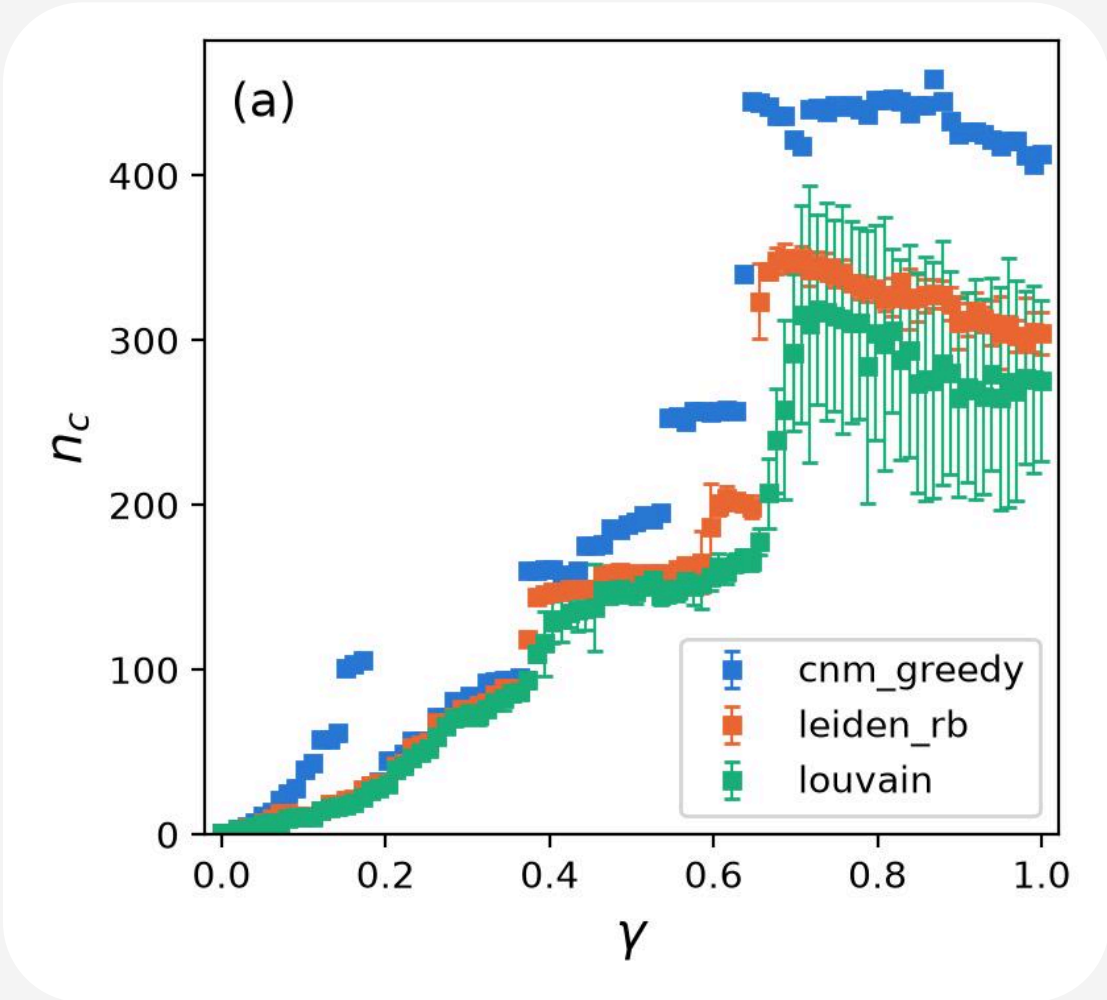


100 GRAPHS
21 CATEGORIES
FROM NETWORKREPOSITORY.COM

JUST 5/100 SHOWED THE
REORGANIZATION BEHAVIOR

(*2) - OPTIMIZATION DEPENDENT

ALGORITHM USED TO OPTIMIZE MODULARITY



MODULARITY

$$Q(G; \gamma) \sum_{g \in G} \frac{1}{2M} \sum_{i, j \in g} \left(A_{ij} - \gamma \frac{k_i k_j}{2M} \right)$$

NULL MODEL COMPARISON

$$Q(G; \gamma) = \frac{1}{M} \sum_g \left(L_g - \gamma \frac{K_g^2}{4M} \right)$$

INTRA-CLUSTERING DENSITY
WITH A REGULARIZATION
OVER VOLUME

SENSITIVITY

$$Q(G;\gamma) = \sum_{g \in G} \frac{1}{2M} \sum_{i,j \in g} \left(A_{ij} - \gamma \frac{k_i k_j}{2M} \right)$$

NULL MODEL COMPARISON

$$Q(G;\gamma) = \frac{1}{M} \sum_g \left(L_g - \gamma \frac{K_g^2}{4M} \right)$$

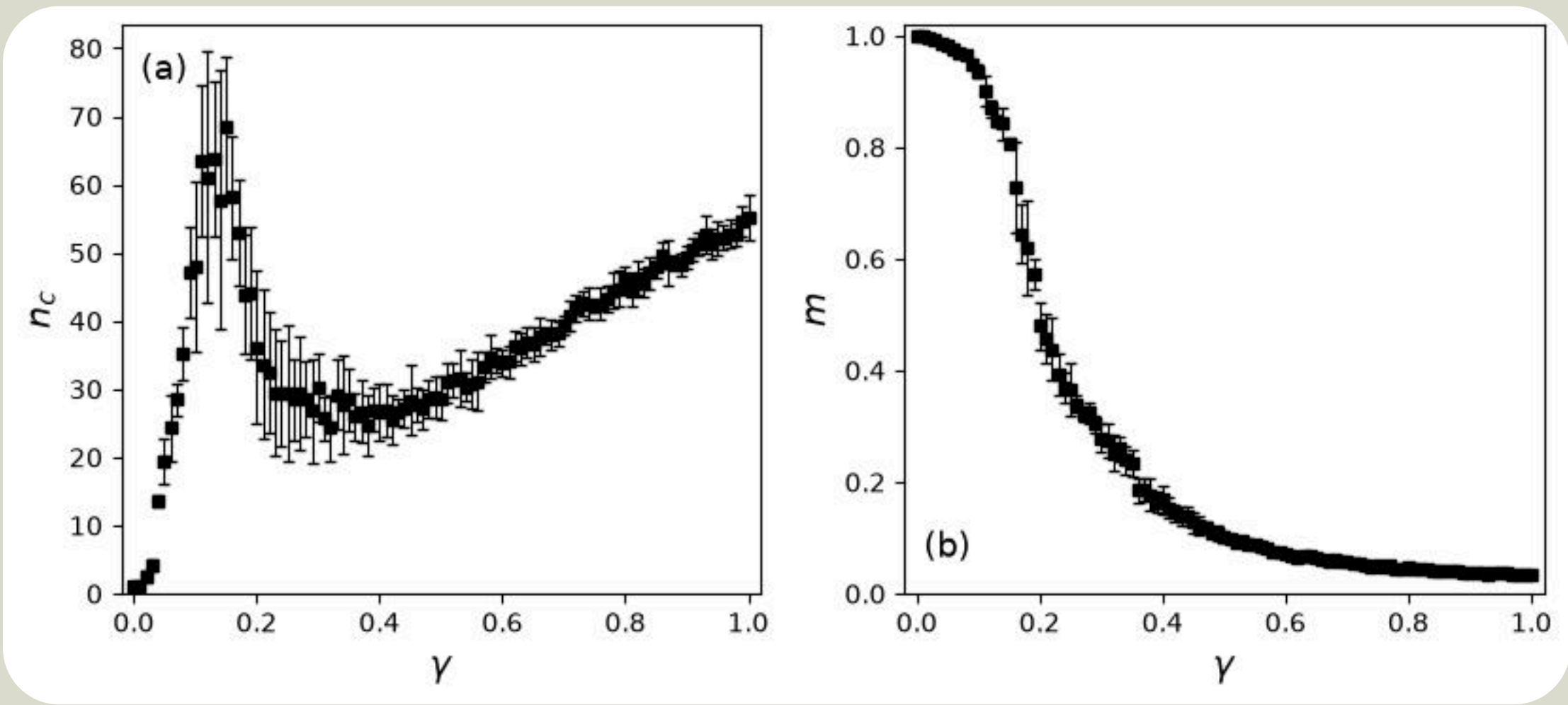
INTRA-CLUSTERING DENSITY
WITH A REGULARIZATION
OVER VOLUME

$$Q(G;\gamma) = q + m\gamma$$

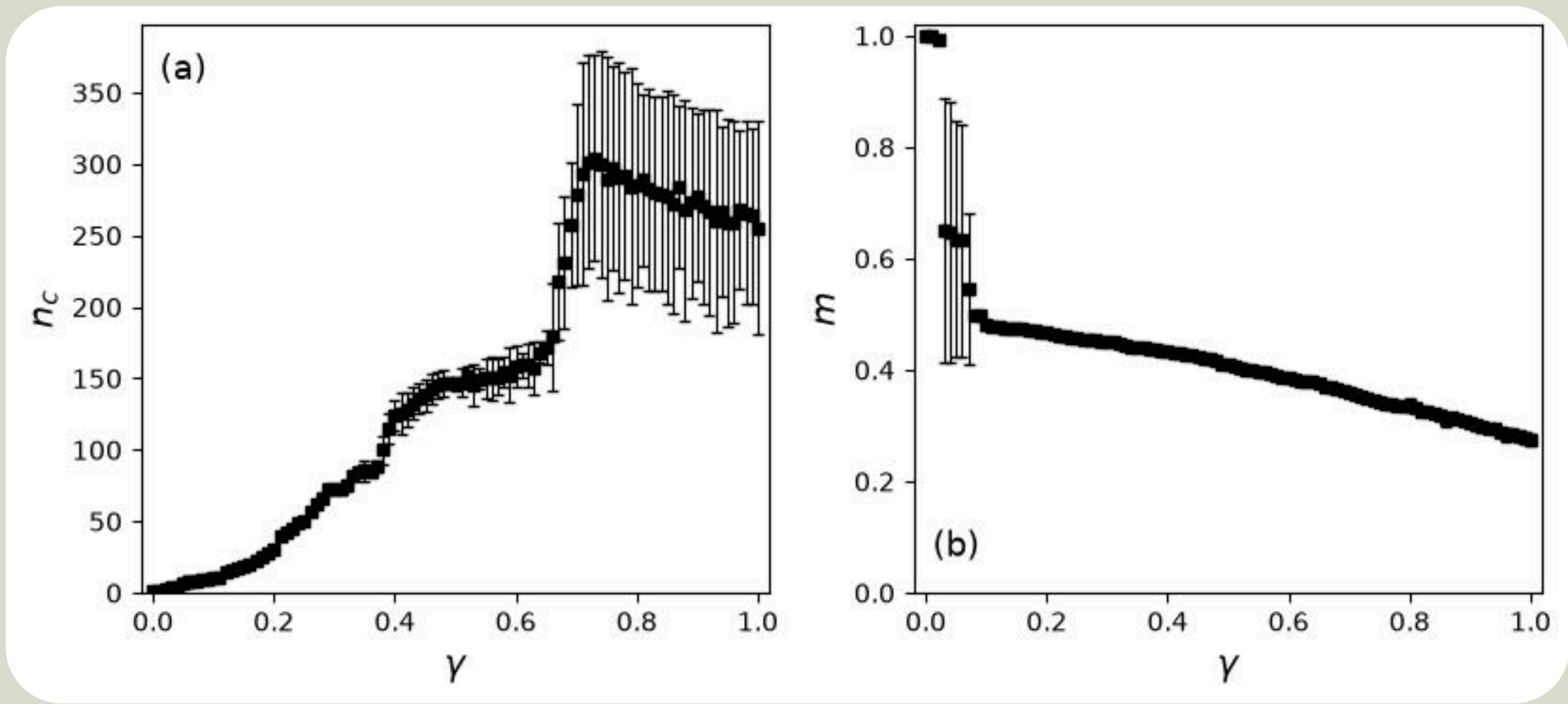
SENSITIVITY

$$Q(G;\gamma) = q + m\gamma$$

CondMat



OpenFlights - routes



MODULARITY

$$Q_{g-h} = \frac{1}{M} \left[L_g + L_h + I_{gh} - \gamma \frac{K_{g-h}^2}{4M} \right]$$

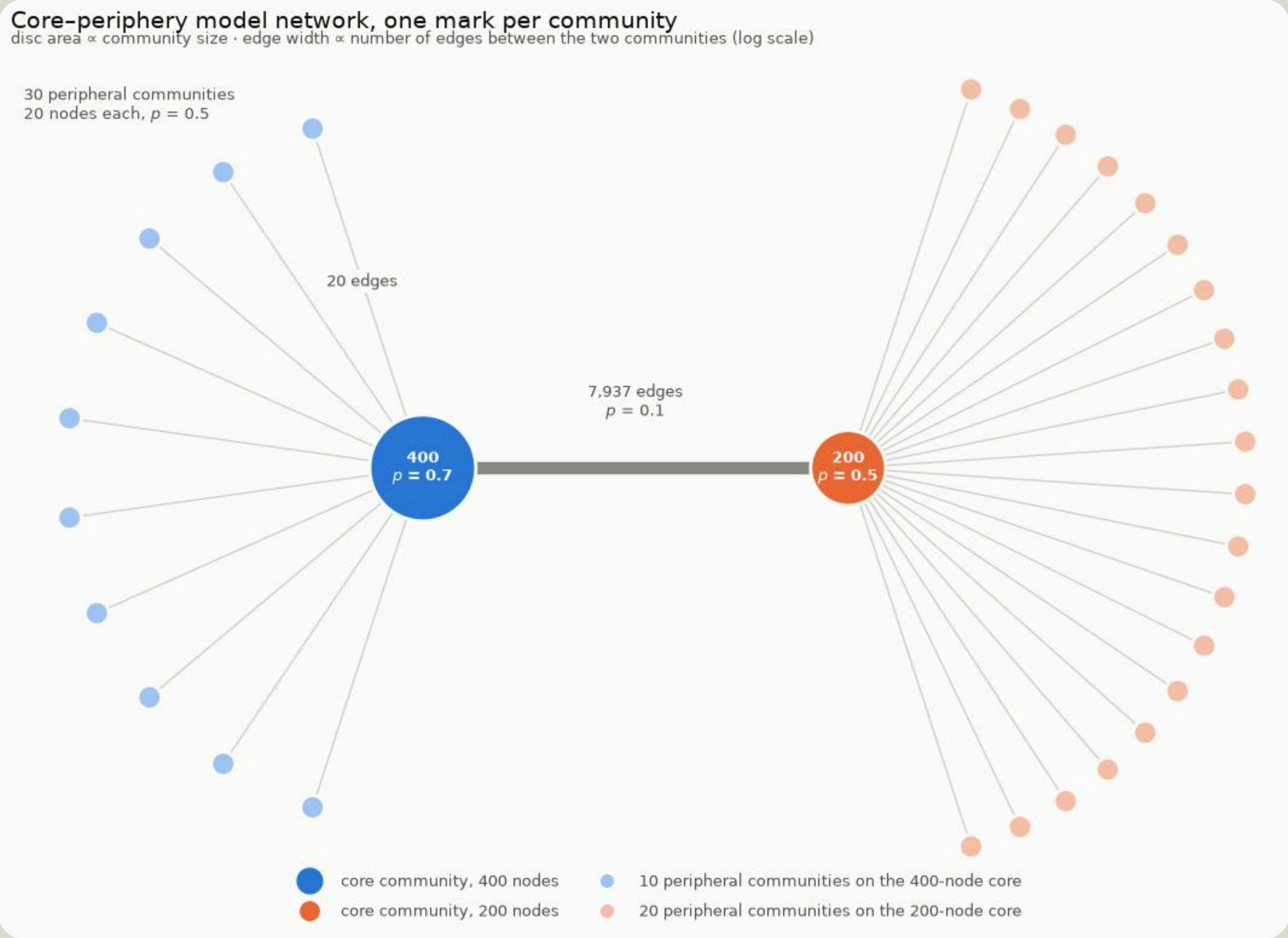
$$Q_g + Q_h = \frac{1}{M} \left[L_g + L_h - \gamma \frac{K_g^2}{4M} - \gamma \frac{K_h^2}{4M} \right]$$

$$\Delta Q(G \rightarrow G'; \gamma) = \frac{I_{gh}}{M} - \gamma \frac{K_g K_h}{2M^2}$$

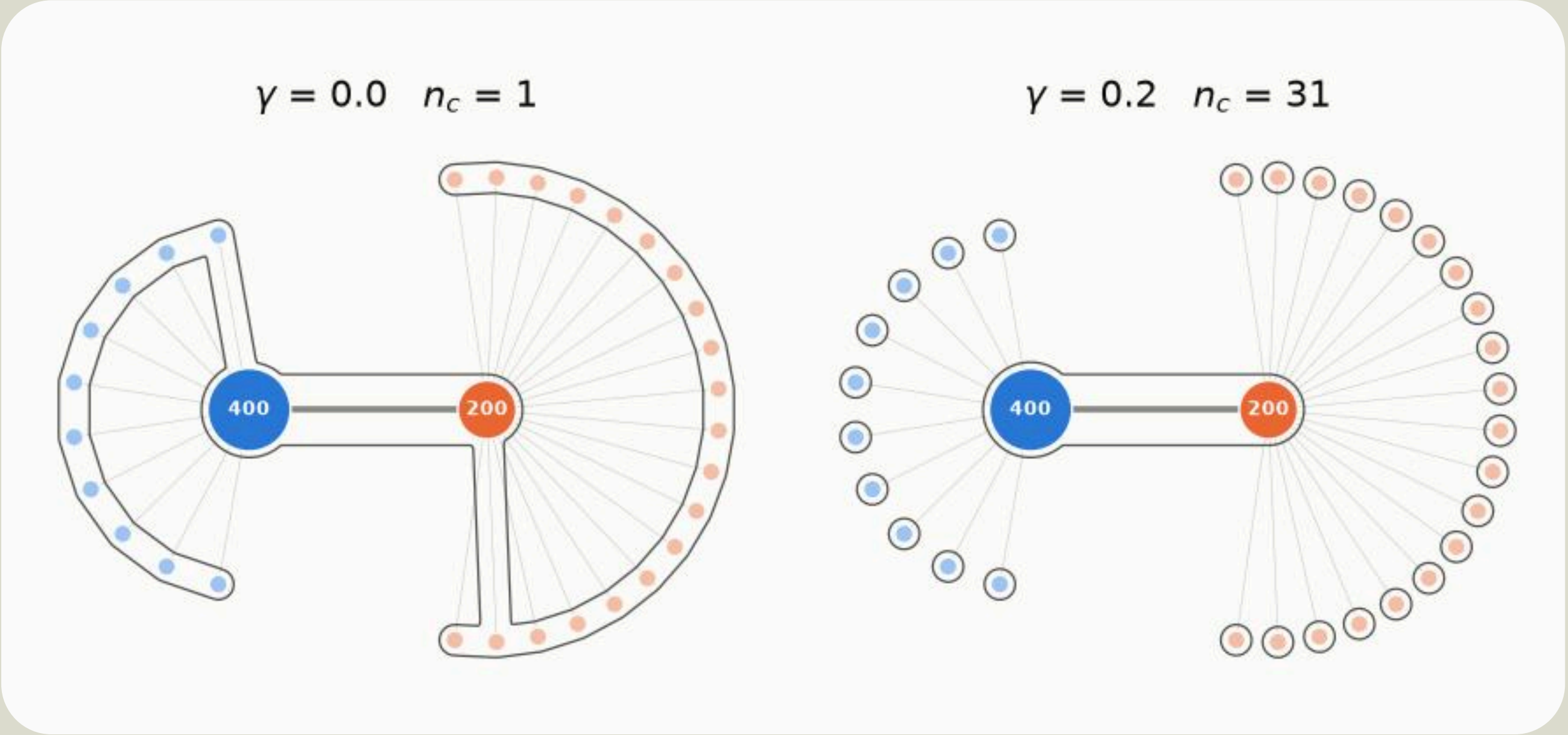
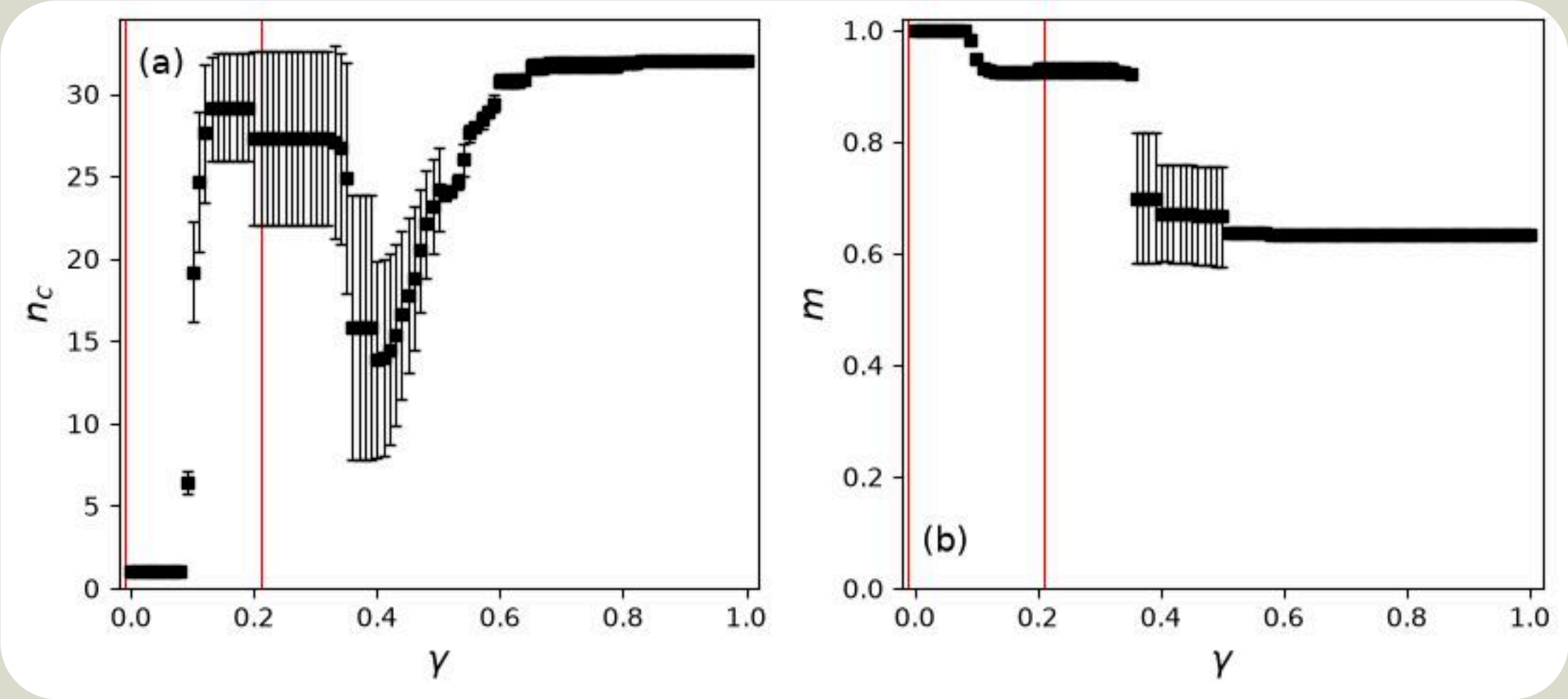
SBM

$$\frac{I_{gh}}{M} - \gamma \frac{K_g K_h}{2M^2}$$

$$q + \gamma m$$



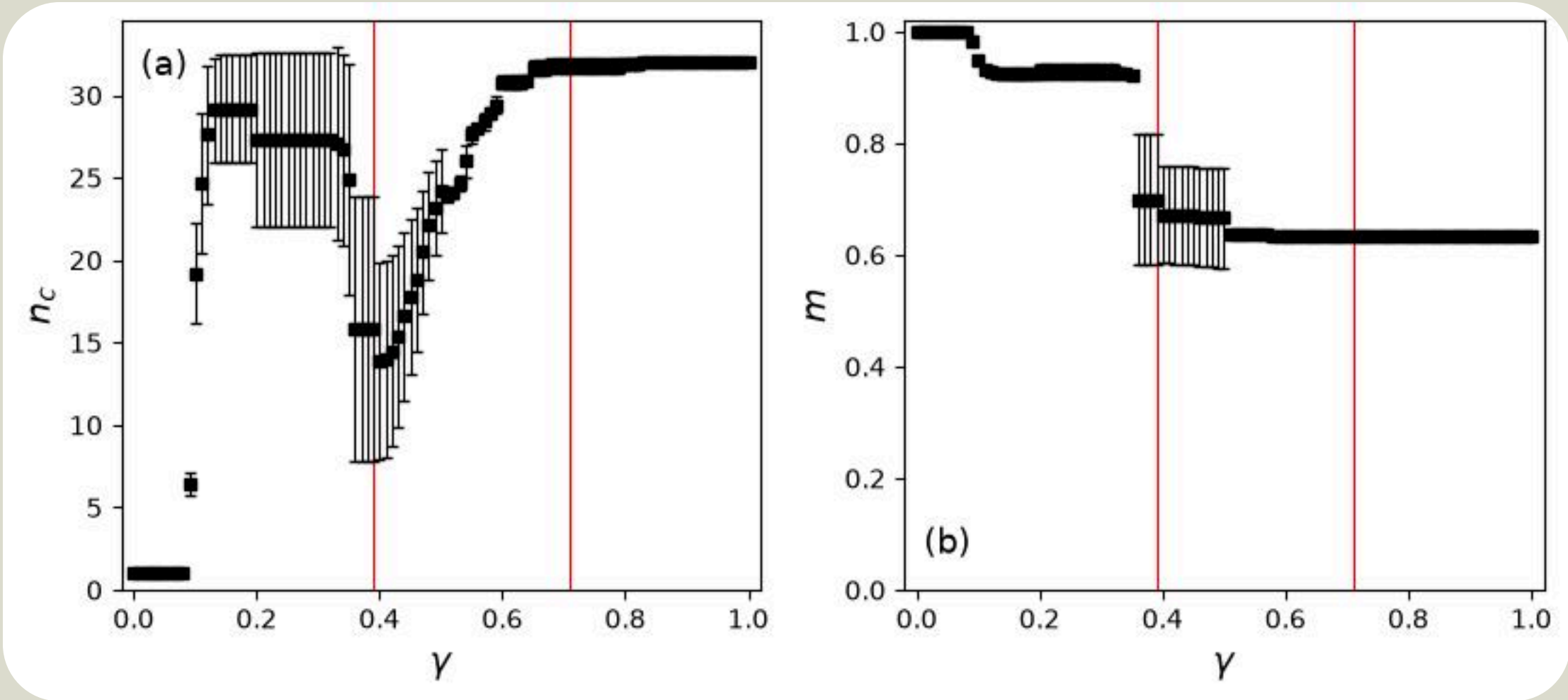
SBM



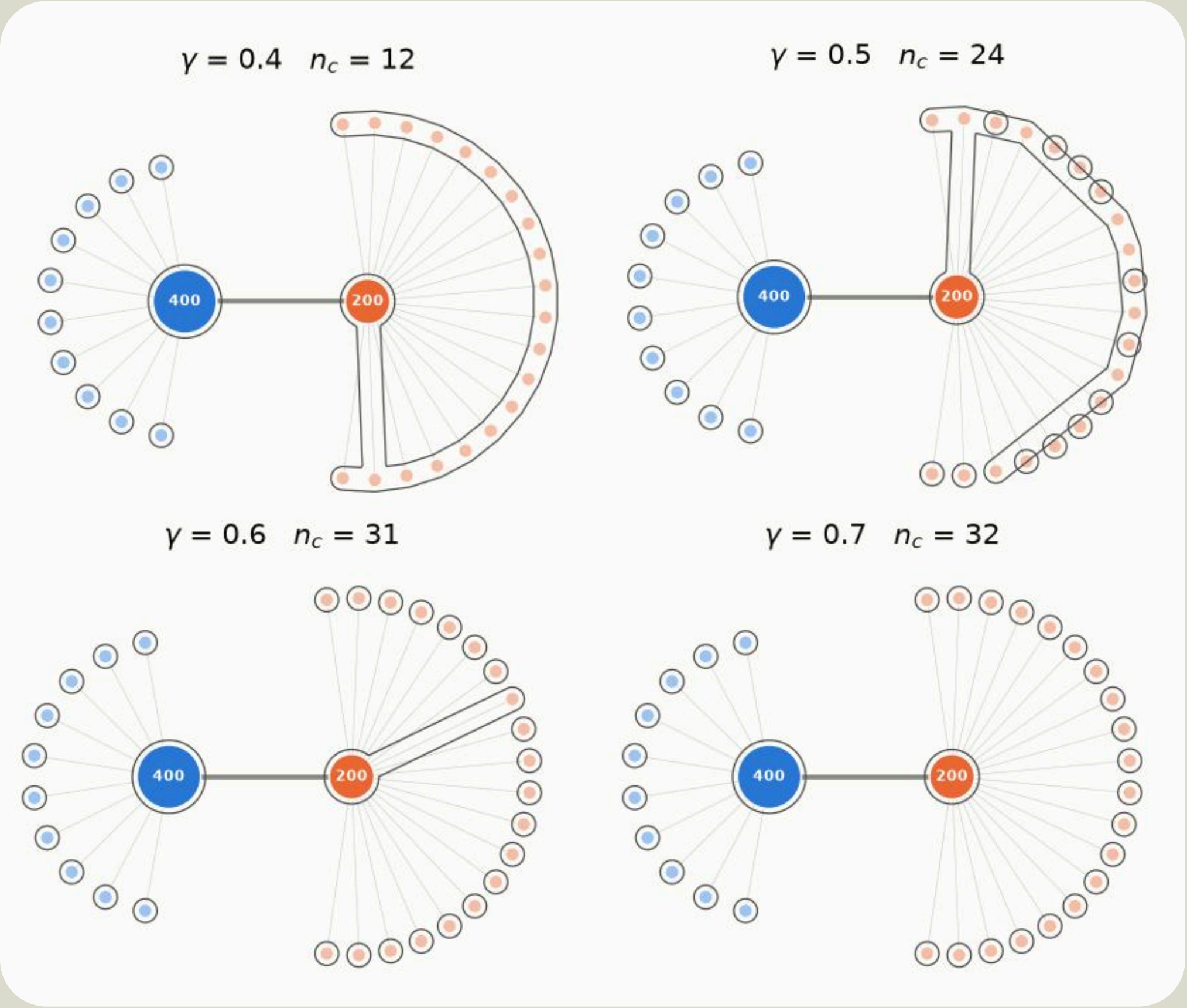
$$q + \gamma m$$

$$\frac{I_{gh}}{M} - \gamma \frac{K_g K_h}{2M^2}$$

SBM



$$q + \gamma m$$
$$\frac{I_{gh}}{M} - \gamma \frac{K_g K_h}{2M^2}$$



THE END

PAPER: [HTTPS://JOURNALS.APS.ORG/PRE/ABSTRACT/10.1103/PHYSREVE.111.014312](https://journals.aps.org/pre/abstract/10.1103/PhysRevE.111.014312)

CODE: [HTTPS://GITHUB.COM/FIGINII/REPLICATION-NETWORK_COMMUNITY_KALEDOSCOPIC_REORGANIZATION](https://github.com/figinii/replication-network_community_kaledoscopic_reorganization)